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COMPUTER ARCHITECTURE (ADVANCED)WK12

1. Performance Analysis Report: HDD vs NVMe SSD

1. Comparison table

The table below compares a modern 7200 RPM SATA hard disk drive (HDD) with a modern NVMe solid-state drive (SSD). The values are typical estimates and may vary depending on the specific model.

Performance metric	7200 RPM SATA HDD	Modern NVMe SSD
Interface	SATA III (6 Gb/s)	PCIe 4.0 x4, NVMe
Sequential read speed	150–220 MB/s	5,000–7,000 MB/s
Sequential write speed	140–210 MB/s	4,000–6,500 MB/s
Random read IOPS	75–150 IOPS	500,000–1,000,000 IOPS
Random write IOPS	75–150 IOPS	500,000–1,000,000 IOPS
Typical access latency	5–15 ms	0.02–0.1 ms

Note: Sequential speeds measure how quickly large blocks of data are transferred. IOPS measures how many input/output operations a device can perform per second. Latency measures the time taken to access data. HDD performance is affected by mechanical movement, while NVMe SSD performance depends on the drive, workload and system configuration.

2. Analysis (approximately 300 words)

Introduction

A Hard Disk Drive (HDD) and a Non-Volatile Memory Express Solid-State Drive (NVMe SSD) are both used to store data in computers. However, they differ in speed, technology and performance. An HDD uses rotating magnetic disks and moving parts, while an NVMe SSD uses flash memory to store data electronically. Manufacturer specifications show that NVMe SSDs can achieve much higher transfer speeds and input/output performance than traditional HDDs.

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Task 1: Booting an operating system

When a computer starts, the operating system loads files, drivers and background services from the storage device into memory. An HDD takes longer because its mechanical parts must move to locate data. Its relatively high latency makes accessing many small files slower.

An NVMe SSD can read many small files much faster because it has no moving parts and offers low latency. Its high random read IOPS also allows it to handle multiple file requests efficiently. As a result, a computer with an NVMe SSD will generally boot faster and become ready to use sooner than a computer with an HDD, although the exact time depends on the operating system and computer hardware.

Task 2: Loading a large video game

Large video games contain many files, including textures, maps, sounds and character models. When loading a game, the computer must read these files from storage.

An HDD transfers data at a much lower speed, which can cause longer loading screens when starting a game or entering a new level. An NVMe SSD can transfer large amounts of data much faster and retrieve many small files efficiently. This can reduce loading times and help game environments load more smoothly. However, actual gameplay frame rates depend mainly on components such as the CPU and GPU, and not only on storage speed.

Conclusion

In conclusion, an NVMe SSD is the better choice for operating system booting and loading large video games because of its higher read/write speeds, greater IOPS and lower latency. An HDD remains useful for storing large amounts of data at a lower cost. Therefore, using an NVMe SSD for the operating system and games, alongside an HDD for extra storage, is a practical solution.

2.Performance Analysis Report: HDD vs NVMe SSD

1. Introduction

Storage devices are used to save data, programs and files on a computer. Two common types are the Hard Disk Drive (HDD) and the NVMe Solid-State Drive (SSD). An HDD uses spinning disks to store data, while an NVMe SSD uses flash memory. This report

compares their performance and explains how they affect computer startup and video game loading.

2. Comparison table

Performance metric	7200 RPM SATA HDD	NVMe SSD (PCIe 4.0)
Interface	SATA III	NVMe over PCIe 4.0
Sequential read speed	150–220 MB/s	5,000–7,000 MB/s
Sequential write speed	140–210 MB/s	4,000–6,500 MB/s
Random read IOPS	75–150 IOPS	500,000–1,000,000 IOPS
Random write IOPS	75–150 IOPS	500,000–1,000,000 IOPS
Latency	5–15 milliseconds	0.02–0.1 milliseconds

Note: These are approximate values. Actual performance depends on the model and computer specifications.

3. Performance analysis (approximately 300 words)

Task 1: Booting an operating system

When a computer starts, the operating system must load files, drivers and other important programs from the storage device into memory. An HDD takes more time because it has moving parts that must spin and move to find the required data. This makes reading many small files slower.

An NVMe SSD is much faster because it has no moving parts and can access data electronically. It also has lower latency and higher IOPS, allowing it to handle many file requests quickly. As a result, a computer using an NVMe SSD usually starts faster and becomes ready to use sooner than a computer using an HDD.

Task 2: Loading a large video game

Large video games contain many files, such as graphics, sounds, maps and character models. These files must be read from storage when the game starts or when a player enters a new level.

An HDD takes longer to load these files because of its slower transfer speeds and mechanical movement. This can result in longer loading screens. An NVMe SSD can read large amounts of data much faster, reducing loading times and helping game

assets load more quickly. However, the actual gaming experience also depends on the computer's processor, RAM and graphics card.

4. Conclusion

In conclusion, an NVMe SSD provides better performance than an HDD because it has higher read/write speeds, more IOPS and lower latency. It is a better choice for booting an operating system and loading large video games. An HDD is still useful for storing large amounts of data at a lower cost. Therefore, using an NVMe SSD for the operating system and games, and an HDD for extra storage, is a good solution.

5. References

1. KIOXIA — SSD product specifications.
2. Hewlett Packard Enterprise — Hard disk drive specifications.